

Unit 9 – Inference for Quantitative Data: Slopes

Notes 1 - Sampling Distributions and Confidence Intervals for Slope

Vitruvius predicted that the length of a man's outstretched arms (armspan) is equal to his height. The linear equation that would predict this relationship is: _____. We could describe this as our population regression line:

$$\mu_y = \alpha + \beta x$$

If we measured the height and armspan of everyone in class, we could create a regression equation, $\hat{y} = a + bx$. Now, if we repeated this activity with a different class, would we get the same least squares regression line? What would be different? What would be the same?

Here is some of this same data, but from other classes. Develop a LSRL for each data set.

Height	Armspan
174	165
168	165
191	185
164	172
159	159
174	175
178	181
155	155
179	176
161	165
171.5	171
179.5	178.5
184	184
182.5	186
168	175
190	179.5
160	160
192	191
189	188
173	166.5
177.5	177.4
177	177.5
183	184.3
183	187
186	180
154	154.5

Height	Armspan
187	182
170	172
170	169
170.5	178
180	175.5
171.5	173.5
182.5	183.5
172	168
161	163
161	161
161	166
185.5	187
177	176.5
172	170
166.5	173
183	186
161	161
177	184
172	172
173	174
172	172
172	168
166	174
178	178
171	177
184	177

Height	Armspan
178	176
165	165.5
191	198
185	176
167.5	175
178	177
179	179
176	180
163	169
166	172
161	164
160	160
189	188
185	182
165	162
171	168
174	184
172	167
179.5	179.5
185	180
174	175
166	160
166	167
182	178
169	165
172	172

How do our sample least square regression lines relate to the population regression line?

If we took many LSRLs of the same size from the sample population, we can create a sampling distribution for our slope:

population regression line: $\mu_y = \alpha + \beta x$

sample regression line: $\hat{y} = a + bx$

Sampling Distribution of the Slope

- Shape: Approximately normal as long as conditions (below) are met
- Mean: $\mu_b = \beta$
- Standard Deviation: $\sigma_b = \frac{\sigma}{\sigma_x \sqrt{n}}$
 - σ = standard deviation of residuals
 - σ_x = standard deviation for explanatory variable
 - n = sample size

Once we develop sampling distribution for our slope, we can begin to ask and answer our inference questions:

- Is there really a linear relationship between x and y in the population, or could the pattern we see in the scatterplot plausibly happen just by chance?
- In the population, how much will the predicted value of y change for each increase of 1 unit in x ? What's the margin of error for this estimate?

Conditions for Regression Inference

- Linear: x and y have a linear relationship
 - Check - make sure scatter plot is linear (or check correlation)
- Independent: If sampling without replacement, check the 10% condition
- Normal: When x is fixed, y follows a normal distribution
 - Check - Make a histogram of the residuals and make sure it looks approximately Normal
- Equal SD: Standard deviation of residuals doesn't change
 - Check - Make a residual plot and check for a random pattern
- Random: Random sampling (SRS) or random assignment (experiment)

Confidence Interval for b

Created to estimate the slope β of the population (true) regression line

$$b \pm t^* \left(\frac{s}{s_x \sqrt{n-1}} \right) \text{ and } t^* \text{ has } C\% \text{ of the area between } -t^* \text{ and } t^* \text{ with } df = n - 2$$

The confidence interval will be interpreted the same way: We are $C\%$ confident that the true slope of the regression line lies between ____ & _____. (Make sure to always include the context of the problem!)

Ex) The data in the following table gives 15 randomly selected large school districts from around the nation, what their average teacher salary is, and how much they spend per pupil (per pupil expenditure). Selected summary statistics are given below as well. Construct and interpret a 95% confidence interval for the slope of the population regression line. What type of relationship does this interval suggest? Explain.

Average Teacher Salary	Per Pupil Expenditure
\$31,418	\$11,443
\$32,226	\$10,589
\$33,483	\$9,809
\$35,474	\$10,205
\$35,923	\$12,645
\$35,943	\$10,045
\$37,636	\$9,075
\$38,181	\$11,968
\$39,236	\$15,337
\$42,240	\$11,989
\$45,147	\$14,848
\$45,589	\$23,091
\$46,954	\$14,322
\$46,992	\$12,143
\$51,443	\$18,920

$$S = 2761.6 \quad s_x = 6154.2 \quad n = 15$$

Computer Output

Most AP problems will not require you to do what we did in the problem above. Most inference questions come with a generic computer output, like what is pictured below:

Predictor	Coef	St Dev	t ratio	P
Constant	2.544	0.134	18.955	0.000
Years	0.164	0.057	2,862	0.005

$s = 1.542$ $R\text{-sq} = 67.5\%$ $R\text{-sq}(\text{adj}) = 66.1\%$

Ex) A study attempted to establish a linear relationship between IQ score and musical aptitude. The following table is a partial printout of the regression analysis and is based on a sample of 20 individuals.

Predictor	Coef	St Dev	t ratio	P
Constant	-22.26	12.94	-1.72	.102
IQ	0.4925	0.1215	4.05	.000

$s = 6.143$ $R\text{-sq} = 47.7\%$ $R\text{-sq}(\text{adj}) = 44.8\%$

What is the 99% confidence interval for the slope of the regression line? Does this suggest a linear relationship?

Ex) A random sample of 10 high school students produced the following results for number of hours of television watched per week and GPA

TV hours	12	21	8	20	16	16	24	0	11	18
GPA	3.1	2.3	3.5	2.5	3.0	2.6	2.1	3.8	2.9	2.6

Determine the least squares regression line and give a 95% confidence interval estimate for the true slope. Assume all conditions have been checked and met.

Calculator - T Interval for Slope

G: LinRegTInt

XList:

YList:

Freq: 1

C-Level:

RegEq: